

Link Budget Report

Summary

#	Component	Signal Power Out (W)	Noise Power Out (W)	Signal Gain (dB)	Noise Gain (dB)	SNR (dB)
1	Signal source Signal starting at transmit side	1	0.0	inf	inf	inf
2	Path loss Path from transmitter to receiver	2.27657e-11	0.0	-106.427	inf	inf
3	Rx antenna gain Gain of the receive antenna	4.54236e-11	0.0	3	inf	inf
4	Noise source Noise at receive antenna	4.54236e-11	1	0.0	inf	-103.427
5	Quantization noise	4.54236e-11	1.0002	0.0	0.000883485	-103.428
6	Sub-band Tuner	2.27118e-11	0.10002	-3.0103	-10	-96.4384

Detailed Report

1. Signal source

name	Signal source
description	Signal starting at transmit side
signal power in (W)	0.0
noise power in (W)	0.0
signal power out (W)	1
noise power out (W)	0.0
snr	inf
noise gain	inf
signal gain	inf

2. Path loss

name	Path loss
description	Path from transmitter to receiver
signal power in (W)	1
noise power in (W)	0.0
signal power out (W)	2.27657e-11
noise power out (W)	0.0
speed of light	2.99792e+08
speed of light units	meters per second
free space path loss formula	$(4.0 * \text{Pi} * \text{frequency} * \text{distance} / \text{speed_of_light})^2$
snr	inf
noise gain	inf
signal gain	2.27657e-11

3. Rx antenna gain

name	Rx antenna gain
description	Gain of the receive antenna
signal power in (W)	2.27657e-11
noise power in (W)	0.0
signal power out (W)	4.54236e-11

noise power out (W)	0.0
snr	inf
noise gain	inf
signal gain	1.99526

4. Noise source

name	Noise source
description	Noise at receive antenna
signal power in (W)	4.54236e-11
noise power in (W)	0.0
signal power out (W)	4.54236e-11
noise power out (W)	1
snr	4.54236e-11
noise gain	inf
signal gain	1

5. Quantization noise

name	Quantization noise
description	
signal power in (W)	4.54236e-11
noise power in (W)	1
signal power out (W)	4.54236e-11
noise power out (W)	1.0002
quantization noise (W)	0.000203451
total bits	12
effective bits	8.67807
headroom db	10
headroom bits	3.32193
snr	4.54144e-11
noise gain	1.0002
signal gain	1

6. Sub-band Tuner

name	Sub-band Tuner
description	
signal power in (W)	4.54236e-11
noise power in (W)	1.0002
signal power out (W)	2.27118e-11
noise power out (W)	0.10002
input lower frequency	1e+07
input upper frequency	2e+07
output lower frequency	1.55e+07
output upper frequency	1.65e+07
signal lower frequency in	1.5e+07
signal upper frequency in	1.7e+07
snr	2.27072e-10
noise gain	0.1
signal gain	0.5